

Edgar Gonzalez Fernandez

Reserva de Sequoia 310,
Fracc. Bosque Sereno, AGS
☎ +52 5547750717
✉ egonzalezcv@gmail.com
📄 github.com/egonzalez1989/
Birth date - May 4 1985



Summary

I am a qualified professional with experience in Machine Learning and Artificial Intelligence development and research, with a strong background in image processing, digital image forensics, and statistical learning. Experienced in development, programming, and data analysis, with proven ability to bridge research and production-oriented ML pipelines, with strong Python skills.

Technical Skills

Image Processing	Image forensics, quality assessment, noise modeling, watermarking, compression algorithms, feature extraction.
ML & DL	Classical ML (SVM, GMM, K-Means, Hierarchical Clustering), CNNs, model evaluation.
Information Security	Symmetric and asymmetric ciphers, hash functions, digital signatures. Public Key Infrastructure and security standards (ANSI, FIPS, IEEE, RSA).
Frameworks	Numpy, Scipy, Scikit-learn, OpenCV, AgentPy, Keras.
Languages	Python, R, Java, C++, C#.
Servers	Basic knowledge of JBoss, Apache Tomcat servers, and Linux management.

Education

- 2018–2023 **PhD in Mathematical Engineering, Statistics and Operations Research**, *Complutense University of Madrid*, Madrid, Spain.
Thesis: Image Tampering Detection Techniques Based on Chromatic Interpolation Algorithms and Sensor Noise.
Supervisor: Dr. Luis Javier García Villaba and Dr. Ana Lucila Sandoval Orozco
Notes: PhD with International Mention
- 2016–2020 **PhD in Computer Science**, *Center for Research and Advanced Studies of the National Polytechnic Institute (CINVESTAV-IPN)*, Mexico City.
Thesis: Zero-knowledge Proof Systems Based on Steiner Triple Systems.
Supervisor: Dr. Guillermo Morales Luna and Dr. Feliú Sagols Troncoso
- 2010–2012 **Master in Science (Mathematics)**, *Center for Research and Advanced Studies of the National Polytechnic Institute (CINVESTAV-IPN)*, Mexico City.
Degree project: Elliptic curve cryptosystems over fields of characteristic 3.
Supervisor Dr. Feliú Sagols Troncoso

2003–2010 **Bachelor's in Applied Mathematics**, *Autonomous University of the State of Hidalgo*, Pachuca, Hidalgo.
Degree project: Classification of groups of small order.
Supervisor Dr. Rafael Villarroel Flores

Languages

Spanish Native

English Advanced

TOEFL iBT: 104 pts, ITP: 617 pts

Work Experience

May 2025 **AI Researcher**, *INFOTEC*, Aguascalientes, Mexico.

- Current
- Research on image forensics, computer vision and multi-agent systems, combining signal-processing features with deep learning models.
 - Multi-agent simulation of urban incidents with Weighted A* path planning (arXiv:2601.13452).
 - Invited speaker and reviewer for international conferences on computer vision and cybersecurity.
 - Supervision of graduate theses in data science and machine learning.

Oct 2022 **Professor**, *Tecnológico de Monterrey*, Monterrey, Mexico.

- Aug 2025
- Taught AI applications in data science, multi-agent systems, analysis of algorithms and theory of computation.
 - Taught mathematical modeling for engineering (calculus, statistics, financial mathematics).
 - Evaluated AI and data-science student projects; contributed to curriculum updates.

Jan 2023 **Research Professor (Titular A)**, *INFOTEC*, Aguascalientes, Mexico.

- Jan 2024
- Led research projects on image processing and image forensics.
 - Taught Digital Image Processing and Project Seminars (M.Sc. in Data Science).
 - Coordinated the M.Sc. in Data Science: academic planning, student supervision and graduation.

Feb 2020 **Deputy Manager of Teaching**, *INFOTEC*, Aguascalientes, Mexico.

- Dec 2022
- Managed academic administration and curriculum design for graduate programs, including their accreditation in the National System of Postgraduate Studies (SNP).

Sept 2017 **Research Assistant**, *Universidad Complutense de Madrid*, Madrid, Spain.

- Nov 2019
- Developed computer vision algorithms for large-scale image and video forensic analysis within Horizon 2020 projects.
 - Designed tampering-detection methods based on colour filter array interpolation, sensor noise and compression traces (published in *Expert Systems with Applications* and *IEEE T-ITS*).
 - Built camera source-identification pipelines with supervised and unsupervised learning (SVM, k-means, hierarchical clustering, Gaussian mixture models).
 - Tools: Python, NumPy, SciPy, OpenCV, TensorFlow.
 - Implemented post-quantum multivariate cryptography (<https://github.com/cinvestavcs-mexicocity/MQCrypto>).

- Mar 2012 **Software Developer**, *Industry and research projects*, Mexico City / Guadalajara,
Dec 2015 Mexico.
- Mercadotecnia Ideas y Tecnología (2014–2015): secure payment software in Java; PKI, DUKPT key management and HSM integration for banking compliance.
 - Insys (2013–2014): identity-management services and mobile apps with location services.
 - Conacyt project ECO-2010-C01-149902 (2012–2013): stochastic models and time-series tests (unit root, co-integration, causality) for financial data.

Selected Publications

- 2026 **González Fernández, E.** A simulation of urban incidents involving pedestrians and vehicles based on Weighted A*. *Arxiv e-prints*, **2026**, 2601.13452
- 2023 **González Fernández, E.**; Sandoval Orozco, A.L.; García Villalba, A.L.; A Multi-channel Approach for Detecting Tampering in Colour Filter Images. *Expert Systems with Applications*, **2023** 120498.
- 2022 **González Fernández, E.**; Sandoval Orozco, A.L.; García Villalba, A.L.; Digital Video Manipulation Detection Technique Based on Compression Algorithms. *IEEE Transactions on Intelligent Transportation Systems*, **2022** 23(3), 2596–2605.
- Armas Vega, E.A.; **González Fernández, E.**; Sandoval Orozco, A.L.; García Villalba, A.L.; Copy-move forgery detection technique based on discrete cosine transform blocks features. *Neural Computing and Applications*, **2021**.
- Armas Vega, E.A., **González Fernández, E.**; Sandoval Orozco, A.L.; García Villalba, L.J.; Image Tampering Detection by Estimating Interpolation Patterns. *Future Generation Computer Systems*, **2020**, 10, pp. 229–237.
- González Fernández, E.**, Morales-Luna, G., Sagols Troncoso, F.; A Zero-Knowledge Proof System with Algebraic Geometry Techniques. *Applied Science*, **2020**, 10, 465.
- Armas Vega E.A., **González Fernández, E.**; Sandoval Orozco A.L.; García Villalba, L.J. Passive Image Forgery Detection Based on the Demosaicing Algorithm and JPEG Compression. *IEEE Access*, **2020**, 8, 11815–11823.
- 2018 **González Fernández, E.**; Sandoval Orozco A.L.; García Villalba, L.J.; Hernandez-Castro, J. Digital Image Tamper Detection Technique Based on Spectrum Analysis of CFA Artifacts, *Sensors*, **2018**, 18(9), 2804.

Memberships

- 2025–2029 Member of the National System of Researchers SNII level I.